

## Syllabus ( ME 231 )

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### Basic information

**Course title:** Statics

**Course code:** ME 231

**Lecturer:** [Assist. Prof. Saeed LOTFAN](#)

**ECTS credits:** 5

**GTU credits:** 3 ( )

**Year, Semester:** 2, Fall

**Level of course:** First Cycle (Undergraduate)

**Type of course:** Compulsory

**Language of instruction:** English

**Mode of delivery:** Face to face

**Pre- and co-requisites:** none

**Professional practice:** No

**Purpose of the course:** This course teaches fundamental engineering science and mathematical principles required for a proper understanding of mechanics. Students will learn to use vector methods and free body diagram development as tools to logically approach and solve engineering mechanics problems. In this course the student will develop engineering problem solving methods through fundamental introductory topics in mechanics including: particle and rigid body equilibrium in 2D and 3D force systems, appropriate support reactions, moments of forces, equivalent systems, distributed forces, center of gravity, trusses, frames, internal forces (including shear and bending moment diagrams), moment of inertia, parallel axis theorem, mass moment of inertia. Upon completion of this course students are expected to understand how to analyze practical engineering structures under force and moment systems and have a strong foundation of the engineering mechanics principles and methods needed for both use as qualified engineers and for secondary courses in mechanics.

## Learning outcomes

*Upon successful completion of this course, students will be able to:*

1. Be able to solve problems which require the ability to analyze systems of forces in static equilibrium.

### Contribution to Program Outcomes

1. Ability to identify, formulate and solve complex engineering problems; ability to select and apply appropriate analysis and modeling methods for this purpose.
2. Ability to select and use modern techniques and tools necessary for the analysis and solution of complex problems encountered in engineering practice; ability to use information technologies effectively.
3. An ability to design and conduct experiments, collect data, analyze and interpret results for the study of complex engineering problems or discipline-specific research topics.
4. Analytical thinking ability, ability to transform ideas into practice, ability to design, ability to think in three dimensions.

### Method of assessment

1. Written exam
  2. Homework assignment
2. Posses ability to work professionally in both thermal and mechanical systems areas including the design and realization of such systems.

### Contribution to Program Outcomes

1. Ability to identify, formulate and solve complex engineering problems; ability to select and apply appropriate analysis and modeling methods for this purpose.
2. Ability to select and use modern techniques and tools necessary for the analysis and solution of complex problems encountered in engineering practice; ability to use information technologies effectively.
3. An ability to design and conduct experiments, collect data, analyze and interpret results for the study of complex engineering problems or discipline-specific research topics.
4. Information about the effects of engineering practices on health, environment and safety in universal and social dimensions and the problems of the age reflected in the field of engineering; awareness of the legal consequences of engineering solutions.
5. Analytical thinking ability, ability to transform ideas into practice, ability to design, ability to think in three dimensions.
6. The ability to work professionally by preparing and managing projects in the fields of mechanical, thermal systems or automatic control.

### Method of assessment

1. Written exam
  2. Homework assignment
3. To become skillful with the mathematical and graphical techniques of vector analysis in mechanical engineering.

### Contribution to Program Outcomes

1. Ability to identify, formulate and solve complex engineering problems; ability to select and apply appropriate analysis and modeling methods for this purpose.
2. Ability to select and use modern techniques and tools necessary for the analysis and solution of complex problems encountered in engineering practice; ability to use information technologies effectively.
3. An ability to design and conduct experiments, collect data, analyze and interpret results for the study of complex engineering problems or discipline-specific research topics.
4. Analytical thinking ability, ability to transform ideas into practice, ability to design, ability to think in three dimensions.

#### Method of assessment

1. Written exam
2. Homework assignment

#### Contents

**Week 1:** Introduction, Syllabus, Review, Units. Fundamental Principles, Newton's Laws / Force Vectors

**Week 2:** Cartesian Vectors and their addition and subtraction, Position Vectors / Force Vectors

**Week 3:** Equilibrium of a Particle / Force System Resultants 1

**Week 4:** Force System Resultants 2 / Equilibrium of Rigid Body in 2D

**Week 5:** Equilibrium of Rigid Body in 3D / Structural Analysis

**Week 6:** Frames and Machines / Internal Forces

**Week 7:** Midterm-1,  
Internal Forces, Shear and Moment Equations and Diagrams

**Week 8:** Cables / Friction

**Week 9:** Applications of Friction / Center of Gravity and Centroids

**Week 10:** Fluid Statics / Moments of Inertia

**Week 11:** Max and Min moments / Mass moments of Inertia

**Week 12:** Midterm-2,  
Mass moments of Inertia

**Week 13:** Virtual Work

**Week 14:** Solving Review Problems

**Week 15\*:** Review

**Week 16\*:** Final Exam

**Textbooks and materials:** R. C. Hibbeler, Engineering Mechanics Statics, Thirteenth Edition, Pearson Prentice Hall. [ISBN-10: 0132915545]

- Recommended readings:**
1. Beer, Johnston, Cornwell, Sanghi; Vector Mechanics for Engineers: Statics and Dynamics, 10th Edition, McGraw Hill.
  2. Anthony Bedford and Wallace Fowler, Engineering Mechanics, Statics and Dynamics, 3rd Edition, Prentice Hall, 2002 (or later editions)
  3. Boresi, A.P., and Schmidt, R.J., Engineering Mechanics: Statics and Dynamics, Brooks/Cole Publishing Company, Boston, 2001.
  4. Hibbeler, R.C., Engineering Mechanics: Statics and Dynamics, Eighth Edition, Prentice-Hall, Inc., New Jersey, 1998.
  5. McGill, D.J., and King, W.W., Engineering Mechanics: Statics and Dynamics, Third Edition, PWS Publishing Company, Boston, 1995.
  6. Merriam, J.L., and Kraige, L.G., Engineering Mechanics: Statics and Dynamics, Fourth Edition, John Wiley & Sons, Inc., New York, 1997.
  7. Pytel, A., and Kiusalaas, J., Engineering Mechanics: Statics and Dynamics, Second Edition, Brooks/Cole Publishing Company, Boston, 1999.
  8. Riley, W.F., and Sturges, L.D., Engineering Mechanics: Statics and Dynamics, Second Edition, John Wiley & Sons, Inc., New York, 1996.

\* Between 15th and 16th weeks is there a free week for students to prepare for final exam.

### Assessment

| Method of assessment   | Week number | Weight (%) |
|------------------------|-------------|------------|
| Mid-terms:             | 7, 12       | 40         |
| Other in-term studies: | -           | -          |
| Project:               | 14          | 10         |
| Homework:              | 1-14        | 10         |
| Quiz:                  | 1-14        | 10         |
| Final exam:            | 16          | 30         |
| Total weight:          |             | 100 (%)    |

### Workload

| Activity                         | Duration (Hours per week) | Total number of weeks | Total hours in term |
|----------------------------------|---------------------------|-----------------------|---------------------|
| Courses (Face-to-face teaching): | 3                         | 14                    | 42                  |
| Own studies outside class:       | 4                         | 14                    | 56                  |
| Practice, Recitation:            | -                         | -                     | -                   |
| Homework:                        | -                         | -                     | -                   |
| Term project:                    | -                         | -                     | -                   |
| Term project presentation:       | -                         | -                     | -                   |
| Quiz:                            | -                         | -                     | -                   |
| Own study for mid-term exam:     | 8                         | 2                     | 16                  |
| Mid-term:                        | 2                         | 2                     | 4                   |
| Personal studies for final exam: | 7                         | 1                     | 7                   |
| Final exam:                      | -                         | -                     | -                   |
| Total workload:                  |                           |                       | 125                 |
| Total ECTS credits:              |                           |                       | 5*                  |

\* ECTS credit is calculated by dividing total workload by 25.

(1 ECTS = 25 work hours)

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| <b>General information:</b> | 2-2-0 = 3 GTU credits, 5 ECTS credits,<br>Th: 09:30-12:20, (Face-to-Face at Harita Amfi 1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Instructor:</b>          | Assist. Prof. Saeed LOTFAN<br>H2 123, slotfan@gtu.edu.tr, (262) 605 26 58                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Teaching assistants:</b> | Dr. İlke ALGÜL<br>H2 Z34, ilkees@gtu.edu.tr, (262) 605 27 99<br>Arş. Gör. Eref Bedirhan AKSOY<br>H2 Z34, bedirhanaksoy@gtu.edu.tr, (262) 605 27 99                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Office hours:</b>        | Th: 14:00-17:00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Textbooks:</b>           | <ul style="list-style-type: none"> <li>Hibbeler, R. C., "Engineering Mechanics Statics", Thirteenth Edition, Pearson Prentice Hall. [ISBN-10: 0132915545]</li> <li>Beer, F.P., Johnston, E.R., "Vector Mechanics for Engineering Statics", McGraw-Hill International Book Company</li> <li>Meriam, J. L., Kraige, L. G., "Engineering Mechanics, Statics", John Wiley&amp;Sons Inc.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Catalog description:</b> | The main purpose of this course is to provide the student with a clear and thorough presentation of the theory and application of engineering mechanics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Prerequisite:</b>        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Course type:</b>         | This course is required for all Mechanical Engineering students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Integrity statement:</b> | All students are expected to act with civility, academic and personal integrity, respect others' dignity, rights, and property; and help create and maintain an environment in which all can succeed through the fruits of their own efforts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Grading:</b>             | <p>Midterms 25% each<br/>Homeworks, Project and Quizzes 20%<br/>Final exam 30%</p> <ol style="list-style-type: none"> <li>Students who fail to meet the <b>70%</b> attendance requirement in the course fail with an <b>NA</b> grade; they are not entitled to take the final exam.</li> <li>A student who fails to achieve at least <b>30%</b> success in the weighted average of midterm exam and quizzes evaluations during the term will fail with a <b>VF</b> grade; they are not entitled to take the final exam.</li> <li>Students whose success in the final exam is below <b>25/100</b> are considered to have failed the course with an <b>FF</b> grade, regardless of their other studies.</li> <li>Not failing the items a, b and c above does not mean to have succeeded in the course; for students who meet the minimum requirements stated in these articles, course grade evaluation is made separately in the [FF-AA] range.</li> </ol> |

**Topics covered and Schedule:**

| Week | Date    | Topic                                                                                        |    |
|------|---------|----------------------------------------------------------------------------------------------|----|
| 1    | Sep. 26 | Introduction, Syllabus, Review, Units. Fundamental Principles, Newton's Laws / Force Vectors |    |
| 2    | Oct. 3  | Cartesian Vectors and their addition and subtraction, Position Vectors / Force Vectors       |    |
| 3    | Oct. 10 | Equilibrium of a Particle / Force System Resultants 1                                        |    |
| 4    | Oct. 17 | Force System Resultants 1 / Equilibrium of Rigid Body in 2D                                  |    |
| 5    | Oct. 24 | Equilibrium of Rigid Body in 3D / Structural Analysis                                        |    |
| 6    | Oct. 31 | Frames and Machines / Internal Forces                                                        |    |
| 7    | Nov. 7  | Internal Forces, Shear and Moment Equations and Diagrams                                     | M1 |
| 8    | Nov. 14 | Cables / Friction                                                                            |    |
| 9    | Nov. 21 | Applications of Friction / Center of Gravity and Centroids                                   |    |
| 10   | Nov. 28 | Build Statics / Moments of Inertia                                                           |    |
| 11   | Dec. 5  | Max and Min moments / Mass moments of Inertia                                                |    |
| 12   | Dec. 12 | Mass moments of Inertia                                                                      | M2 |
| 13   | Dec. 19 | Virtual Work                                                                                 |    |
| 14   | Dec. 26 | Solving Review Problems                                                                      |    |
| 15   | Jan. 2  | Review                                                                                       |    |
| 16   | Jan. 9  | Final Exam                                                                                   | F  |

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| <b>General information:</b> | 2-2-0 = 3 GTU credits, 5 ECTS credits,<br>Th: 09:30-12:20, (Face-to-Face at Elect- Z02)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Instructor:</b>          | Assist. Prof. Saeed LOTFAN<br>H2 123, slotfan@gtu.edu.tr, (262) 605 26 58                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Teaching assistants:</b> | Muhammet Bilal İĞDİ<br>H2 Z34, bilaligdi@gtu.edu.tr, (262) 605 27 99<br>Eda ÖZYILMAZ<br>H2 Z16, eozyilmaz@gtu.edu.tr, (262) 605 27 89                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Office hours:</b>        | Th: 14:00-17:00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Prerequisite:</b>        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course type:</b>         | This course is required for all Mechanical Engineering students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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**Topics covered and Schedule:**

| Week | Date    | Topic                                                                                        |    |
|------|---------|----------------------------------------------------------------------------------------------|----|
| 1    | Oct. 5  | Introduction, Syllabus, Review, Units. Fundamental Principles, Newton's Laws / Force Vectors |    |
| 2    | Oct. 12 | Cartesian Vectors and their addition and subtraction, Position Vectors, Force Vectors        |    |
| 3    | Oct. 19 | Equilibrium of a Particle, Free body diagrams, Coplanar force systems                        |    |
| 4    | Oct. 26 | Three-Dimensional Force Systems                                                              |    |
| 5    | Nov. 2  | Three-Dimensional Force Systems -Continue Moment of a Force (Vector)                         |    |
| 6    | Nov. 9  | Moment of a Force (Scalar), Cross Product                                                    |    |
| 7    | Nov. 16 | Internal Forces, Shear and Moment Equations and Diagrams                                     | M1 |
| 8    | Nov. 23 | Relations Between Shear and Moment                                                           |    |
| 9    | Nov. 30 | Center of Gravity, Center of Mass and the Centroid of a Body                                 |    |
| 10   | Dec. 7  | Equilibrium of a Rigid Body FBD's, Equations of Equilibrium Two- and Three-Force Members     |    |
| 11   | Dec. 14 | 3-D FBD's, Equilibrium Equations, Constrains and Statical Determinacy                        |    |
| 12   | Dec. 21 | Simple Trusses, The Method of Joints Sections                                                | M2 |
| 13   | Dec. 28 | Zero-Force Members                                                                           |    |
| 14   | Jan. 4  | Frames and Machines, Moments of Inertia for Areas                                            |    |
| 15   | Jan. 11 | Review                                                                                       |    |
| 16   | Jan. 18 | Final Exam                                                                                   | F  |

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|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| <b>Instructor:</b>          | Assist. Prof. Saeed LOTFAN<br>H2 123, slotfan@gtu.edu.tr, (262) 605 26 58                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Teaching assistants:</b> | Onur Kaya<br>H2 Z16, onur.kaya@gtu.edu.tr, (262) 605 27 89<br>Erdem Baydır<br>H2 Z34, erdembaydir@gtu.edu.tr, (262) 605 2799                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Office hours:</b>        | M: 9:00-12:00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Grading:</b>             | <p>Midterms            25% each<br/>Homeworks        10%<br/>Final exam         40%</p> <ol style="list-style-type: none"> <li>Students who fail to meet the <b>70%</b> attendance requirement in the course fail with an <b>NA</b> grade; they are not entitled to take the final exam.</li> <li>A student who fails to achieve at least <b>30%</b> success in the weighted average of midterm exam and quizzes evaluations during the term will fail with a <b>VF</b> grade; they are not entitled to take the final exam.</li> <li>Students whose success in the final exam is below <b>30/100</b> are considered to have failed the course with an <b>FF</b> grade, regardless of their other studies.</li> <li>Not failing the items a, b and c above does not mean to have succeeded in the course; for students who meet the minimum requirements stated in these articles, course grade evaluation is made separately in the [FF-AA] range.</li> </ol> |

**Topics covered and Schedule:**

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| 4    |      | Three-Dimensional Force Systems                                                              |    |
| 5    |      | Three-Dimensional Force Systems -Continue<br>Moment of a Force (Vector)                      |    |
| 6    |      | Moment of a Force (Scalar), Cross Product                                                    |    |
| 7    |      | Internal Forces, Shear and Moment Equations and Diagrams                                     | M1 |
| 8    |      | Relations Between Shear and Moment                                                           |    |
| 9    |      | Center of Gravity, Center of Mass and the Centroid of a Body                                 |    |
| 10   |      | Equilibrium of a Rigid Body FBD's, Equations of Equilibrium Two- and Three-Force Members     |    |
| 11   |      | 3-D FBD's, Equilibrium Equations, Constrains and Statical Determinacy                        |    |
| 12   |      | Simple Trusses, The Method of Joints Sections                                                | M2 |
| 13   |      | Zero-Force Members                                                                           |    |
| 14   |      | Frames and Machines, Moments of Inertia for Areas                                            |    |
| 15   |      | Review                                                                                       |    |
| 16   |      | Final Exam                                                                                   | F  |